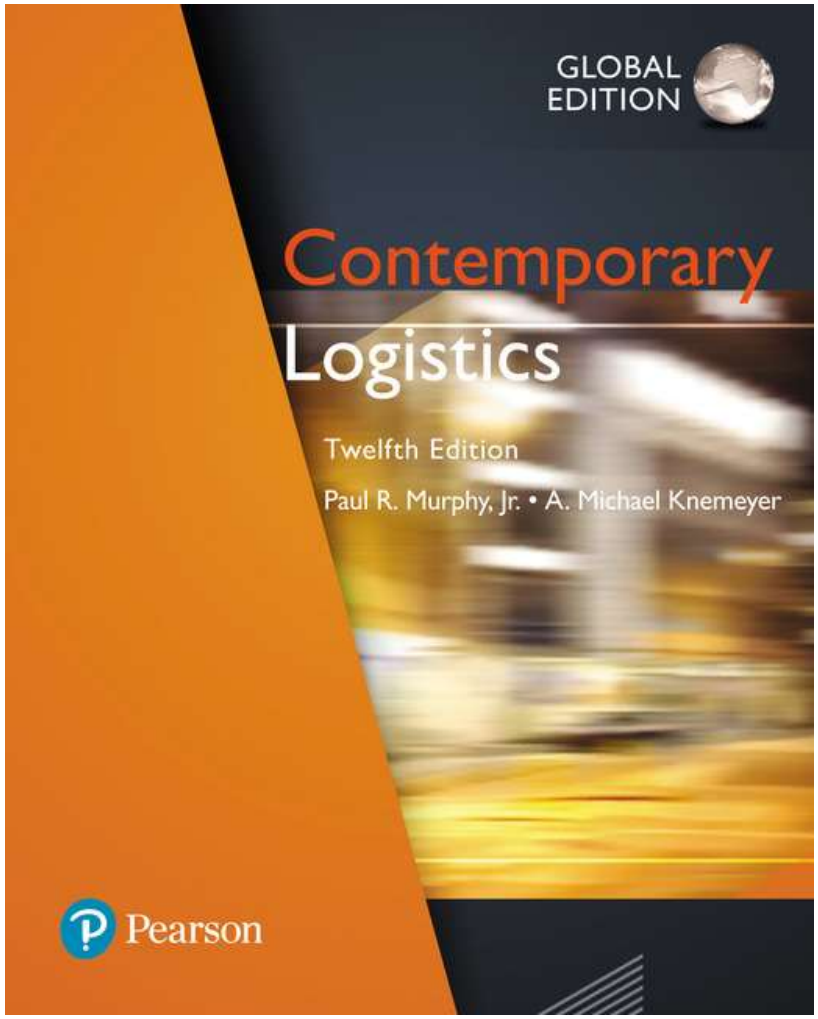
Contemporary Logistics

Twelfth Edition, Global Edition



Chapter 10

Warehousing Management

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Learning Objectives

10.1 To discuss the role of warehousing in a logistics system

10.2 To learn about public, private, contract, and multiclient warehousing

10.3 To analyze select considerations when designing warehousing facilities

10.4 To examine some prominent operational issues in warehousing

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

The Role of Warehousing in a Logistics System (1 of 8)

- Warehousing
  - Refers to "that part of the firm's logistics system that stores products (raw materials, parts, goods-in-process, finished goods) at and between points of origin and point of consumption"[1]
- Warehousing and transportation are substitutes for each other, with warehousing having been referred to as "transportation at zero miles per hour"

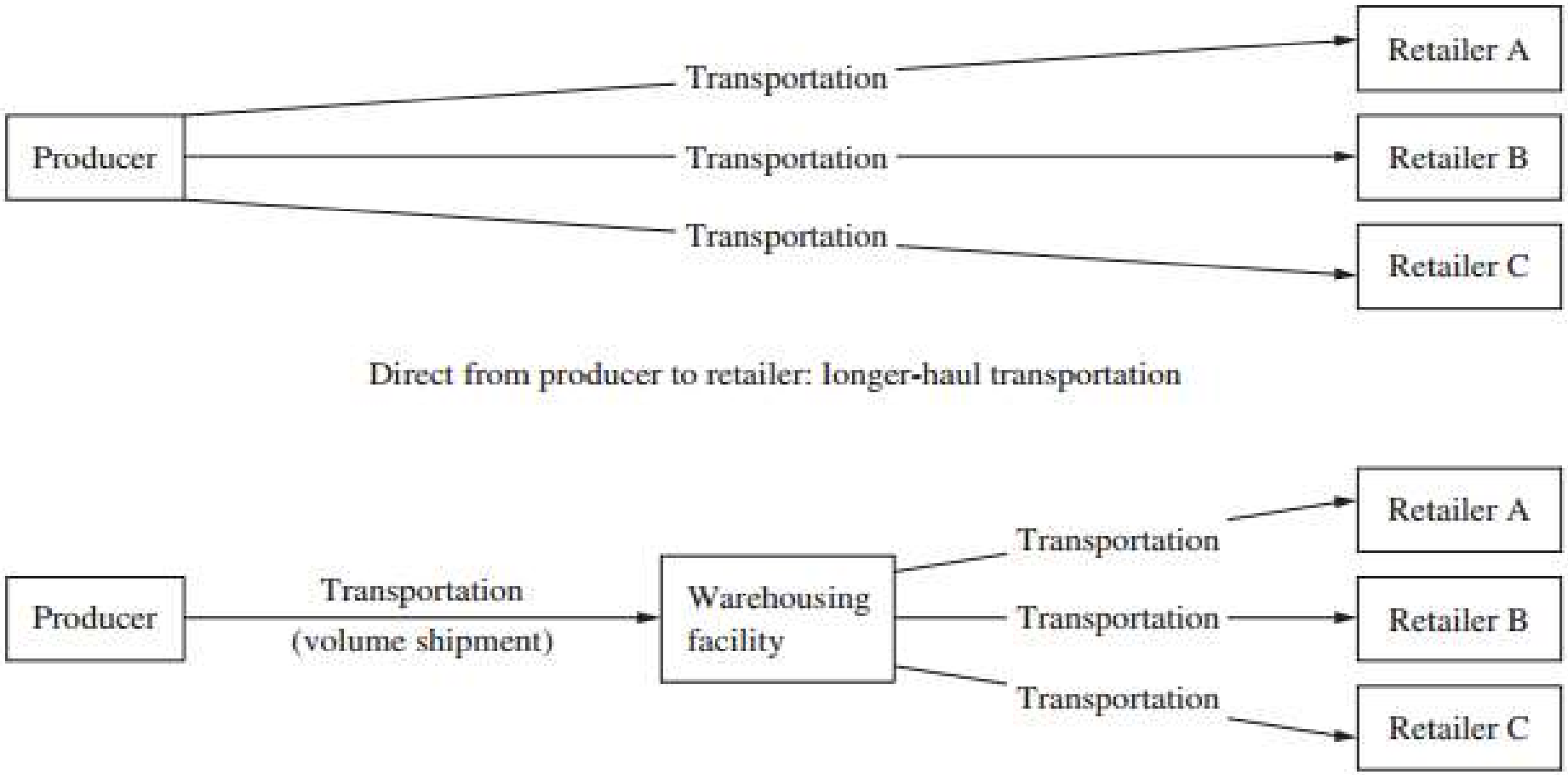
(Instead of moving goods, we keep them in one place until needed.)

[1]Douglas M. Lambert, James R. Stock, and Lisa M. Ellram, *Fundamentals of Logistics Management* (New York: Irwin McGraw-Hill, 1998), Chapter 8.

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Figure 10.1: Adding a Warehousing Facility: Shorter-Haul Transportation



Long-haul transportation vs short-haul transportation

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

The Role of Warehousing in a Logistics System (3 of 8)

- Serves to match different rates or volumes of flow when patterns of production and consumption do not coincide. Production and consumption don't always happen at the same time.
- larger quantities of goods are purchased than can be consumed in a short period of time, and warehousing space is needed to store the surplus product
- Facilitates the **regrouping function** in a supply chain
  - Involves rearranging the quantities and assortment (mix) of products as they move through the supply chain
  - 4 forms of regrouping: **accumulating, allocating, assorting, and sorting out**

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

The Role of Warehousing in a Logistics System (4 of 8)

- Regrouping can take place in four forms:
  - Accumulating (bulk making)
    - Involves bringing together similar stocks from different sources
  - Allocating (bulk breaking)
    - Involves breaking larger quantities into smaller quantities
  - Assorting
    - Refers to building up a variety of different products for resale to particular customers
  - Sorting out
    - "Separating products into grades and qualities desired by different target markets"[2]

[2]William D. Perreault, Jr., Joseph P. Cannon, and E. Jerome McCarthy, *Basic Marketing*, 16th ed. (New York: McGraw-Hill Irwin, 2008), Chapter 11.

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

The Role of Warehousing in a Logistics System (5 of 8)

- Warehousing can be provided by:
  - Warehouses: owned, rented, public, central, cool, quarantine, spare parts
  - Distribution centers: Central DC, and Regional DC
  - Fulfillment centers: warehouse dedicated for e-commerce
  - Cross-docking facilities
  - Depot (raw materials and semi-finished products)

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

The Role of Warehousing in a Logistics System (6 of 8)

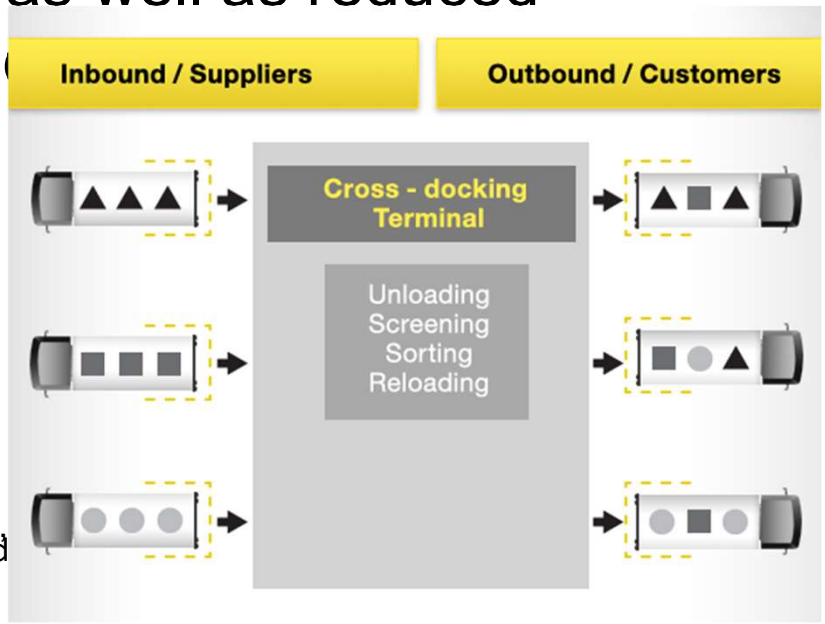
- Warehouses
  - Emphasize the storage of products
  - Primary purpose is to maximize the usage of available storage space
- Distribution centers
  - Emphasize rapid movement of products through the facility
  - Attempt to maximize throughput defined as the amount of product entering and leaving a facility in a given time period

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

The Role of Warehousing in a Logistics System (7 of 8)

- Fulfillment centers
  - Focused on e-commerce (online shopping) orders
  - Amazon Fulfillment Center Video Tour - YouTube
- Cross-docking
  - Defined as "the process of receiving product and shipping it out the same day or overnight without putting it into storage"[3]
  - Key benefits include improved service by allowing products to reach their destinations more quickly as well as reduced inventory carrying costs from less sto product delivery[4]
  - Has grown due to the increased emphasis on time reduction in supply chains. Example video



[3]No author, "2008 Cross-Docking Trends Report," Saddle Creek Corporation,
[4]Mike DelBoro, "Cross-Docking Rediscovered," *Material Handling & Logistics*,

Pearson
Copyright © 2018 Pearson Education, Ltd

The Role of Warehousing in a Logistics System (8 of 8)

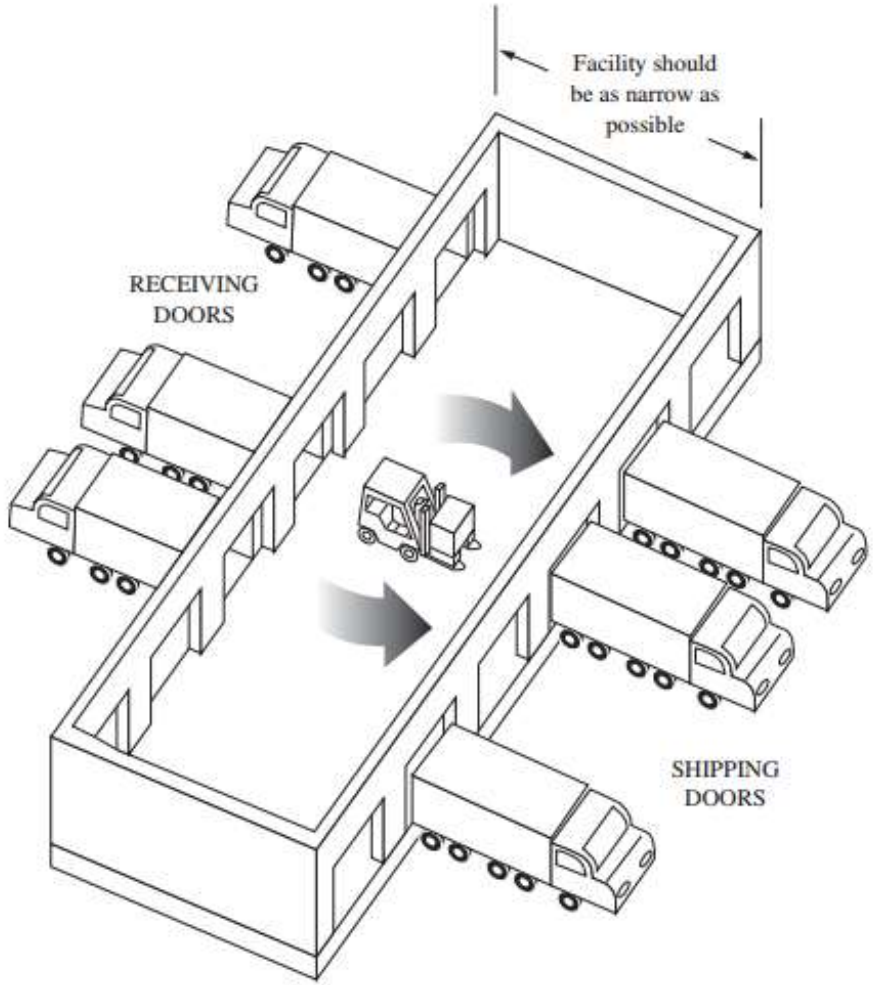
- Cross-docking facilities
  - Is differentiated from distribution centers by the length of time a product is in a facility (24 hours or less for cross-docking facility)
  - Design of the facility is an important consideration to facilitate quick movement of product
    - Should be designed with a minimal amount of storage space and truck doors on two or more sides[5]
    - Depending on the spatial configuration of the land used to build the cross-dock as well a some designs include “H,” “L,” “T,” “U,” and “E” configurations

[5]DelBoro, “Cross-Docking Rediscovered.”

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Figure 10.2: Ideal Facility for Pure Supplier Consolidation (Full Pallet Movement)



I-shaped cross docking design

Source: Reprinted by permission of the Warehousing Education and Research Council.

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Public, Private, Contract, and Multiclient Warehousing (1 of 9)

- Organizations must also decide the proper mix in terms of warehouse ownership
  - Owning
    - Private warehousing
  - Renting (no capital investment on the user's part)
    - Public warehousing
    - Contract warehousing
    - Multiclient warehousing

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Public, Private, Contract, and Multiclient Warehousing (2 of 9)

- Public warehouses characteristics:
  - Serve many users
  - Require no capital investment on the user's part
  - Allow users to rent space as needed. Can be rented on a month-to-month basis
  - Warehousing companies have responsibility for personnel decisions and regulatory issues
    - Warehousing labor safety practices monitored by Occupational Safety and Health Administration (OSHA)
  - Offer more locational flexibility. May provide specialized services
  - Potential drawback is **lack of control** by the user

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Public, Private, Contract, and Multiclient Warehousing (4 of 9)

- Private warehousing
  - Owned by the firm storing goods in the facility
  - Generates high fixed costs
  - Should only be considered by companies dealing with large volumes of inventory
  - Largest users of private warehousing are retail chain stores
  - Offers control to owner
  - Assumes both sufficient demand volume and stability so that warehouse remains full

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Public, Private, Contract, and Multiclient Warehousing (5 of 9)

- Private warehousing
  - Potential drawbacks include:
    - High fixed cost of private storage
    - Necessity of having high and steady demand volumes
    - Less attractive when interest rates are high
    - May reduce an organization's flexibility
      - Internally
      - Externally

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Public, Private, Contract, and Multiclient Warehousing (6 of 9)

- Contract warehousing
  - Also referred to as third-party (3PL) warehousing or dedicated warehousing
  - Refers to "a long term, mutually beneficial arrangement which provides unique and specially tailored warehousing and logistics services exclusively to one client, where the vendor and client share the risks associated with the operation"[6]

[6]Warehousing Education and Research Council, *Contract Warehousing: How It Works and How to Make It Work Effectively* (Oak Brook, IL:
Author, 1993).

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Public, Private, Contract, and Multiclient Warehousing (7 of 9)

- Contract warehousing
  - Less costly than private warehousing and more costly than public warehousing
  - Allows a company to focus on its core competencies
  - Potentially offers the same degree of control as private warehousing due to contract specifications
  - More flexible than private warehousing but less so than public warehousing (depends in part on the length of the contract).

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Public, Private, Contract, and Multiclient Warehousing (8 of 9)

- Multiclient warehousing
  - Mixes attributes of contract and public warehouses
  - Services are more differentiated than those in a public facility
  - Services are less customized than those in contract warehousing
  - Services are purchased through minimum one-year contracts

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Public, Private, Contract, and Multiclient Warehousing (9 of 9)

- Multiclient warehousing
  - Are attractive to smaller organizations
    - That don’t have sufficient volumes to build their own storage facilities
    - That don’t have sufficient volume to justify using contract warehousing
    - Have regular, but not continuous, need for specialized equipment or services
    - Can benefit from working with other companies that have similar needs or requirements

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Warehouse Design

Design Considerations in Warehousing (1 of 8)

- General considerations
  - Quantity and character of goods to be handled must be known
  - Know the purpose to be served by a particular facility
    - Facility with low rates of product turnover should be laid out in a manner that maximizes utilization of the cubic capacity of the storage facility
    - Facility that emphasizes rapid product movement with limited time in storage should be configured to facilitate the flow of products into and out of it

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Design Considerations in Warehousing (2 of 8)

- Trade-offs
  - Must be made among space, labor, and mechanization with respect to warehouse design

Conventional racking

Mobile racking

Example

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Design Considerations in Warehousing (3 of 8)

- Fixed versus variable slot locations for merchandise
  - In fixed slot locations each SKU has one or more permanent slots assigned to it (**velocity slotting**)
  - Variable slot location involves empty storage slots being assigned to incoming products based on space availability

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Design Considerations in Warehousing (4 of 8)

- Build out (horizontal) versus build up (vertical)
  - General rule of thumb is that it is cheaper to build up than build out
  - Warehousing equipment costs tend to increase in vertical

Racking system (vertical)

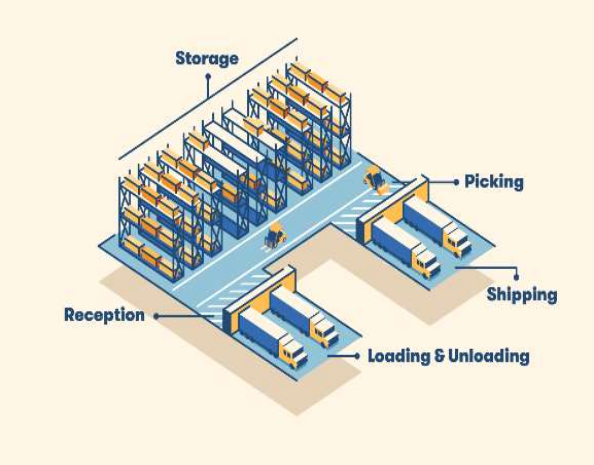
Stocking steel slab stacks (Horizontally)

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Design Considerations in Warehousing (5 of 8)

- Order-picking versus stock-replenishing functions
  - Organizations must decide whether workers who pick outgoing orders and those who are restocking storage facilities should work at the same time or in the same area
- Two-dock versus single-dock layout
  - Generally receiving docks on one side of a facility and shipping docks on the other side, with goods moving between them
  - In a one-dock system, each and every dock can be used for both shipping and receiving

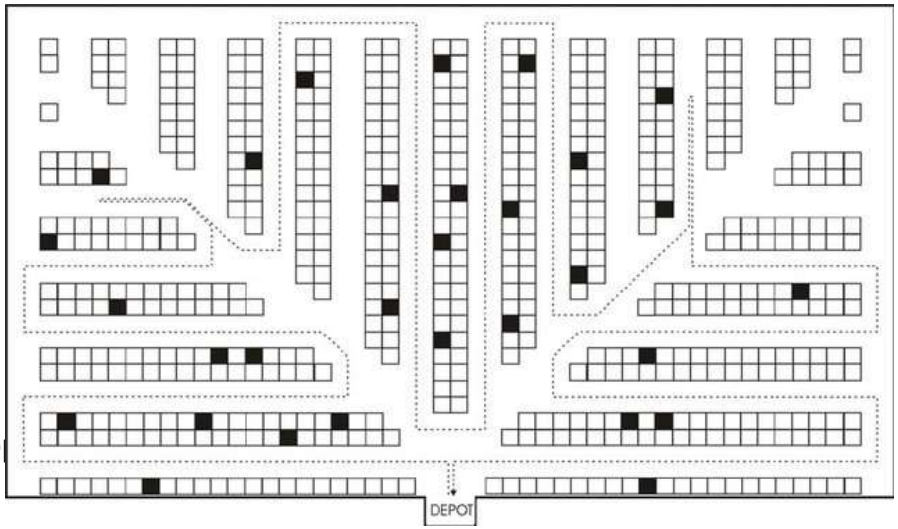


Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Design Considerations in Warehousing (6 of 8)

- Conventional, narrow, or very narrow aisles
  - As aisle space increases, storage capacity decreases
  - Easier to operate mechanical equipment in wider aisles
  - Reduce the chances of accidents and product damage with wider aisles
  - Narrower aisles require specialized storage and handling equipment (more expensive)



Aisles in fishbone warehouse design

Co

Design Considerations in Warehousing (7 of 8)

- Degree of warehouse automation
  - Refers to utilizing mechanical or electronic devices to substitute for human labor
  - Examples include narrow aisle forklifts, automated guided vehicles, automated storage and retrieval systems, radio frequency identification, and others
  - Offers the potential to reduce labor costs and improve warehouse productivity
  - Managers must ensure that automation results in noticeable improvement in warehousing effectiveness and efficiency

Inside A Warehouse Where Thousands Of Robots Pack Groceries

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Design Considerations in Warehousing (8 of 8)

- Other space needs
  - Not uncommon for only approximately 20 percent of a facility's cubic capacity to be occupied by product[7]
  - Warehousing facilities set aside space for nonstorage activities such as employee washrooms, lunchrooms, office space, specialized storage for hazardous items, warehousing supplies, and many more

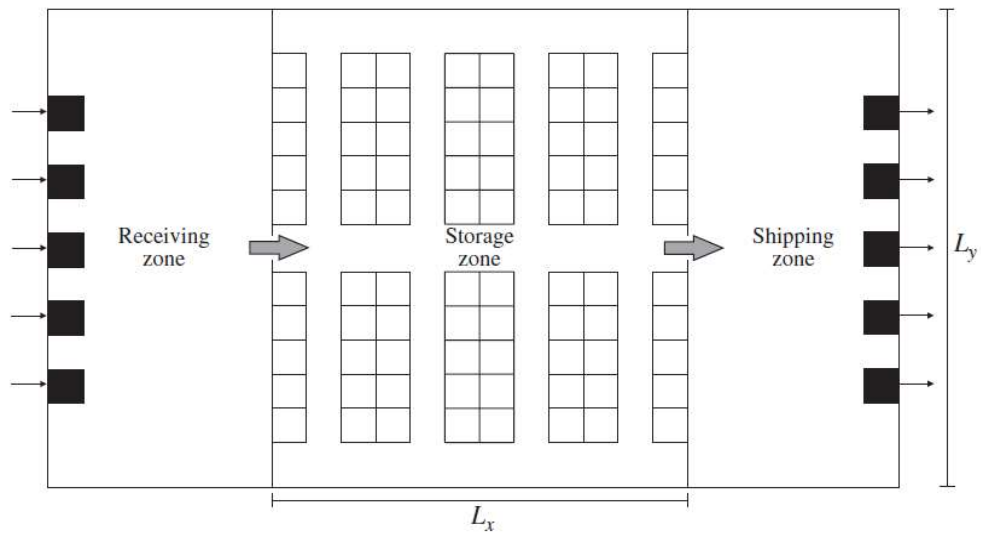
[7]http://www.ioptimizerealty.com/2012/12/21/measuring-utilization-warehouse/

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Sizing of storage zone

Flow through layout

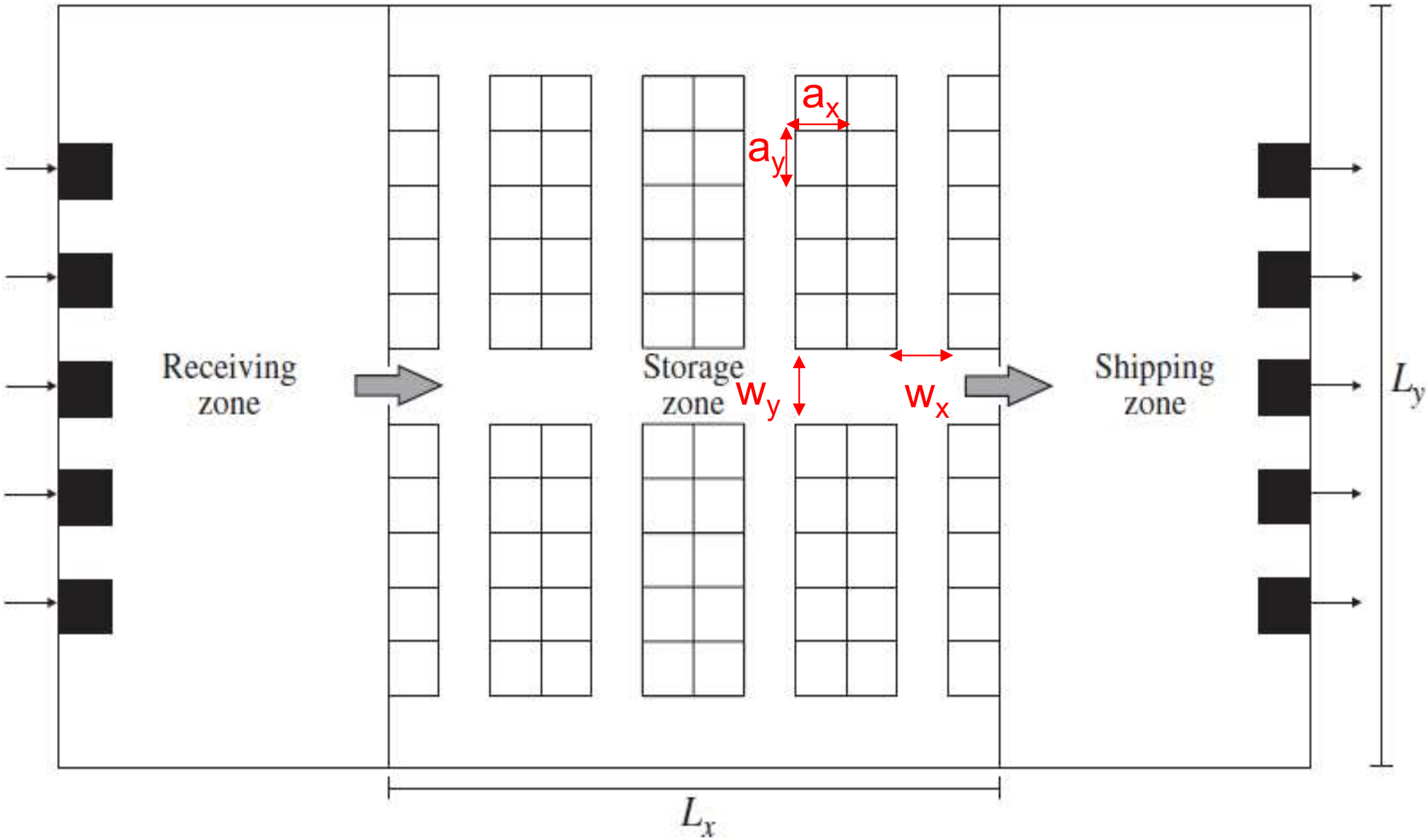


- Determine length L_x and width L_y for given height and required number of storage m
- α_x and α_y unit load (pallet) length and width
- w_x width of vertical aisles, w_y width of horizontal aisles
- n_z is number of locations along z direction
- v the average speed of a picker
- Find n_x number of locations along x direction (even number) and n_y number of locations along y direction. Note,

$$L_x = \left(\alpha_x + \frac{w_x}{2}\right) n_x, \;\; L_y = \alpha_y n_y + 3w_y$$

Pearson
Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Sizing of storage zone

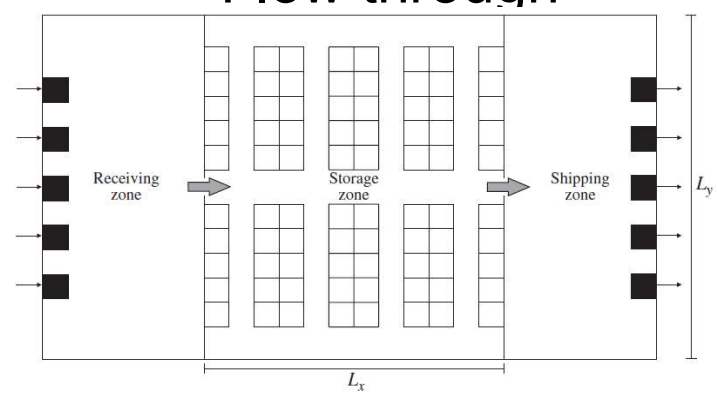


Pearson
Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Sizing of storage zone

$$L_x = \left(\alpha_x + \frac{w_x}{2}\right) n_x, \;\; L_y = \alpha_y n_y + 3w_y$$

Flow through



- Under the hypothesis that a material-handling operation consists of storing or retrieving a single load unit, and all storage locations have the same probability of being accessed, the average distance covered by a picker is (L_x+ L_y/2)

$$\text{Minimize} \;\; \left(\alpha_x + \frac{w_x}{2}\right)\frac{n_x}{v} + \frac{\alpha_y n_y + 3w_y}{2v}$$

subject to

$$n_x n_y n_z \geq m$$

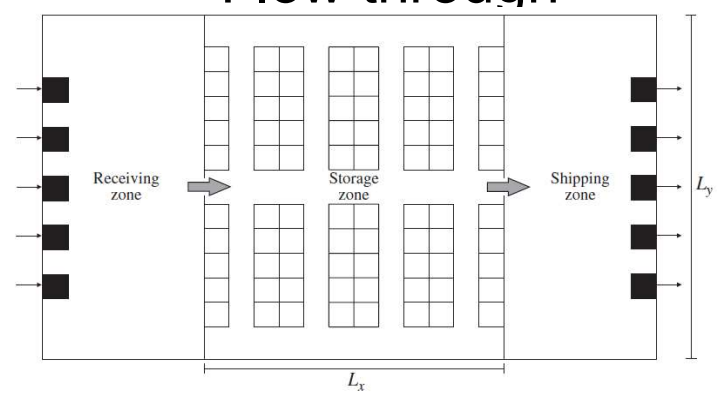
$$n_x, \;\; n_y \geq 0, \;\; \text{integer}$$

$$n_x \;\; \text{even},$$

Pearson
Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Sizing of storage zone

Flow through



Relaxing the integrality constraint:

$$n_x = \frac{m}{n_y n_z}.$$

$$\text{Minimize } \left(\alpha_x + \frac{w_x}{2}\right)\frac{m}{n_y n_z v} + \frac{\alpha_y n_y + 3w_y}{2v}$$

subject to

$$n_y \geq 0.$$

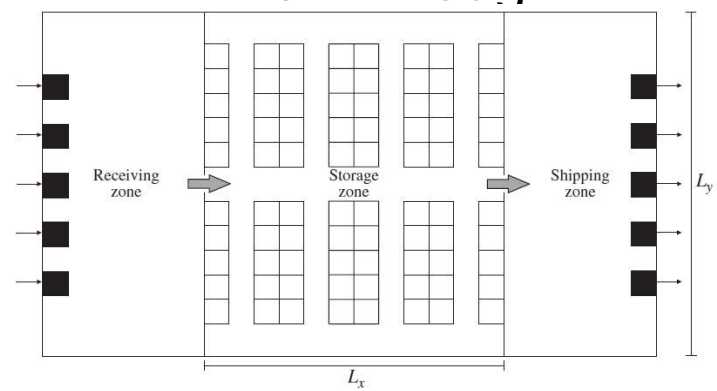
Objective function is convex, the min n_y can be found as follows :

$$\frac{d}{d\,n_y}\left(\left(\alpha_x + \frac{w_x}{2}\right)\frac{m}{n_y n_z v} + \frac{\alpha_y n_y + 3w_y}{2v}\right)\Bigg|_{n_y = n'_y} = 0.$$

Pearson
Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Sizing of storage zone

Flow through



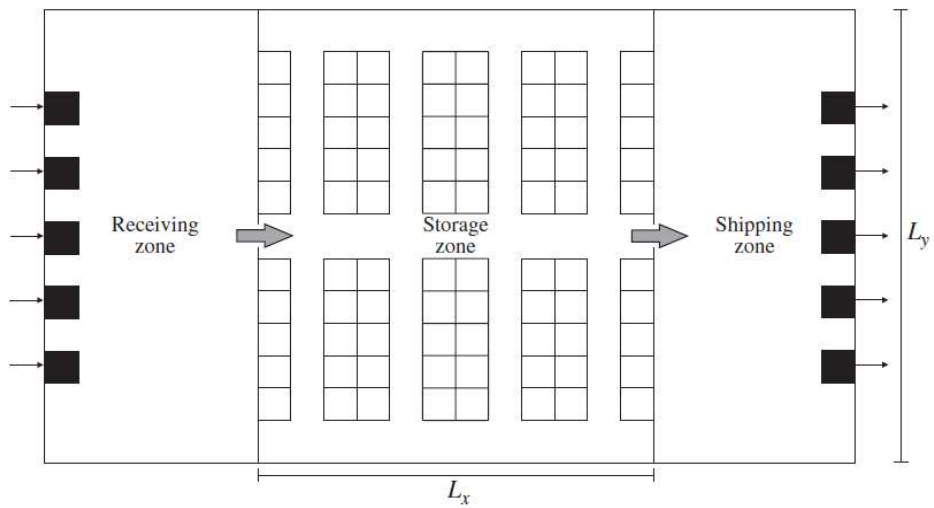
- Minimizing average travel distance of picking unit load gives

$$n_x = \sqrt{\frac{m\alpha_y}{2n_z(\alpha_x + w_x/2)}}, n_y = \sqrt{\frac{2m(\alpha_x + w_x/2)}{\alpha_y n_z}}$$

Pearson
Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Example: storage zone sizing

Wagner Bros plans to build a new warehouse near Sidney, Australia, in order to supply its sales points in New South Wales. On the basis of a preliminary analysis of the problem, it has been decided that the facility should accommodate at least 780 pallets of $90 \times 90\,\text{cm}^2$ dimension. The pallets will be stored onto conventional racks and transported by means of reach trucks. Each rack has four shelves, each of which can store a single pallet. Each pallet occupies a $1.05 \times 1.05\,\text{m}^2$ area. Racks are arranged as in Figure 5.24, where side aisles are 3.5 m wide, while the central aisle is 4 m wide. The average speed of a reach truck is 5 km/h.



Pearson
Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Example: storage zone sizing

- A facility should accommodate at least 780 pallets of 90-by-90cm^2
- Pallets are stored on racks of FOUR shelves
- A pallet occupy 1.05-by-1.05 m^2
- Vertical aisle has 3.5m width and horizontal aisle has 4m width
- Minimizing the total travel time gives

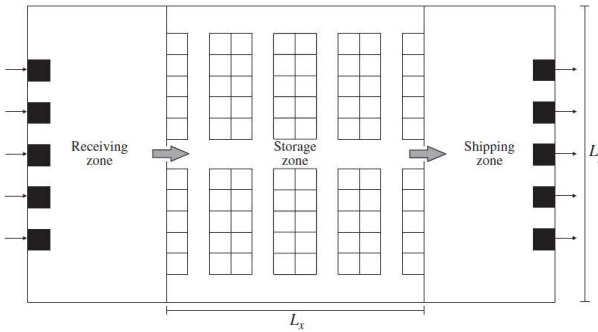
$$n_x = \sqrt{\frac{780 \cdot 1.05}{2 \cdot 4 \cdot \left(1.05 + \frac{3.5}{2}\right)}} = 6.05 \approx 6 \Rightarrow L_x = \left(1.05 + \frac{3.5}{2}\right) \cdot 6 = 16.8m$$

$$n_y = \sqrt{\frac{2 \cdot 780 \cdot \left(1.05 + \frac{3.5}{2}\right)}{4 \cdot 1.05}} = 32.25 \approx 33 \Rightarrow L_y = 1.05 \cdot 33 + 12 = 46.65m$$

What is the total number of storage locations?

Pearson
Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Sizing receiving and shipping zones: Flow through layout (1 Out 2)



- d is average weekly number pallets received
- t is average time needed to unload a vehicle
- q is average number of pallets per vehicle
- T is total time available for unloading operations per week
- The (least) number of docking doors in receiving zone p is

$$p = \left\lceil \frac{d \cdot t}{q \cdot T} \right\rceil$$

- Example: Per week, 21,465 pallets are received in the warehouse. The average time to unload a vehicle is 18.3mins. Vehicle unloading operations has 6.5 hours per day for 6 working days a week. On average, there are 21.7 pallets per vehicle. Find the number docking doors.

$$p = \left\lceil \frac{21465 \cdot 18.3}{21.7 \cdot (6.5 \cdot 6 \cdot 60)} \right\rceil = \lceil 7.8 \rceil = 8\ doors$$

Sizing receiving and shipping zones: Flow through layout (2 Out 2)

- Door width is about 2.6m wide and 1.5m is needed between doors
- An estimate of 50m^2 is needed per unloading door to allow for easy unloading of vehicles
- We find that the length R_x and width R_y of the receiving zone is

$$R_y = 2.6p + 1.5(p+1), \qquad R_x = \frac{50p}{R_y}$$

- Example: A receiving zone is about 400m^2. Find p, R_x and R_y.

$$p = \frac{400}{50} = 8,\ R_y = 2.6 \cdot 8 + 1.5(8+1) = 34.3m, R_x = \frac{400}{34.3} = 11.66m$$

Optimal product allocation to storage locations: tactical decision

- n is the **number of products**
- m_j is the **number of storage** locations required for **product *j*** $(= 1, \ldots, n)$
- k is a **storage location** and in total there are m_d locations
- c_{jk} is the cost to store product j in location k (e.g., mn)
- **Decisions:** whether to store product j in location k

  x_{jk}: 1 if product j is stored in location k, 0 otherwise
- **Objective**: Minimize total cost

$$Minimize \ \sum_{j=1}^{n} \sum_{k=1}^{m_d} c_{jk} x_{jk}$$

- **Constraints**

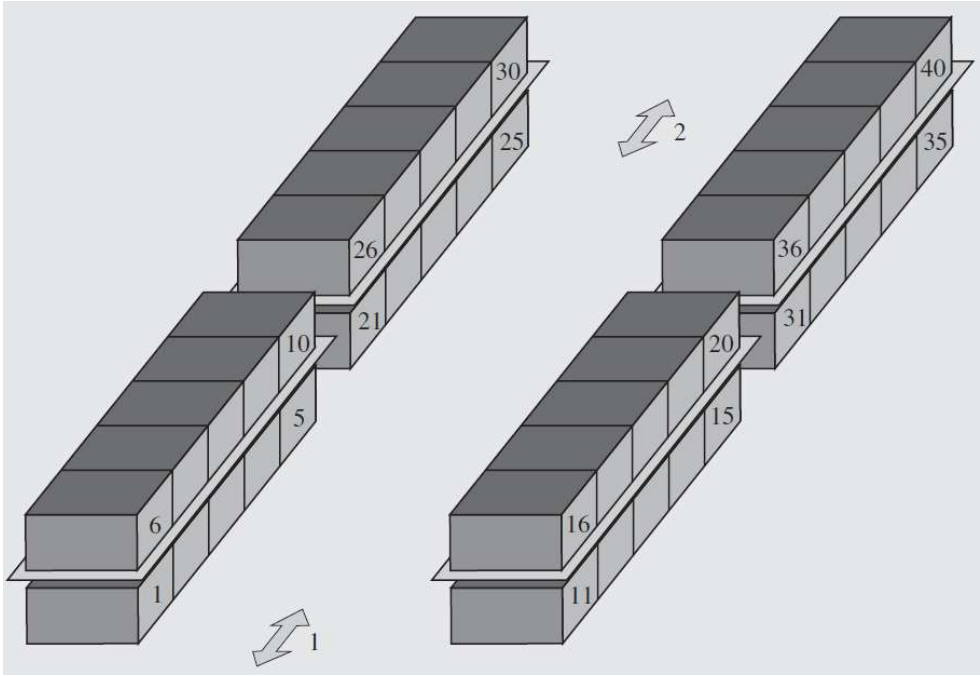
m_j locations are allocated to product j: $\sum_{k=1}^{m_d} x_{jk} = m_j \, , \forall \, j$

location is allocated to at most one product: $\sum_{j=1}^{n} x_{jk} \leq 1, \forall \, k$

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Product allocation Case 1 out 2



- A warehouse has 40 storage locations arranged in 4 conventional racks
- There are 5 products to store with required storage given

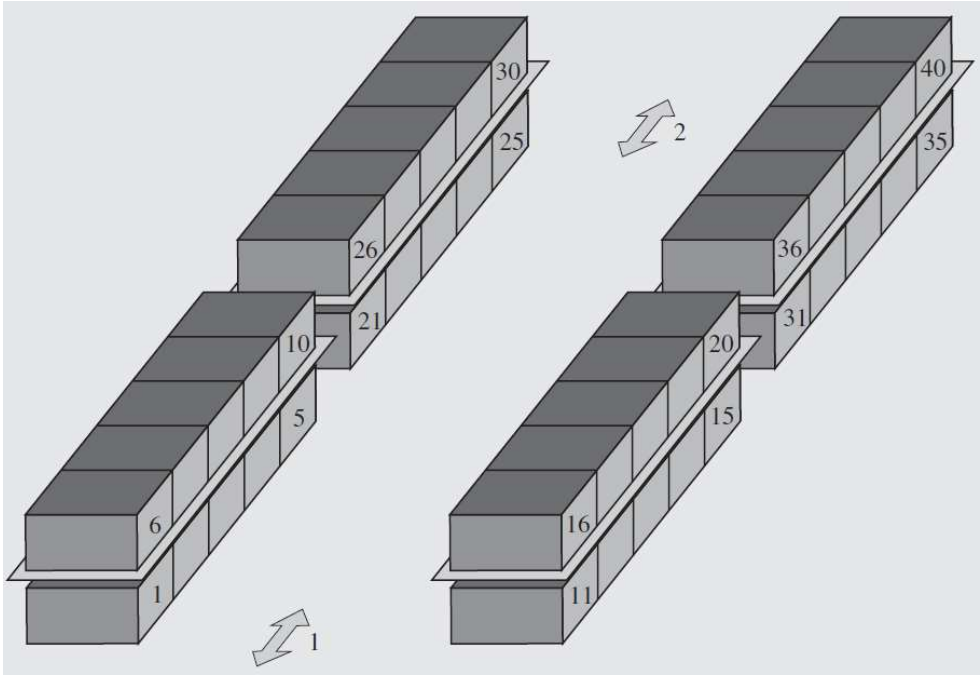
12	6	8	4	8

- The cost coefficients are given (see ProductallocationStorage.xslx)

Storage location	Product 1	Product 2	Product 3	Product 4	Product 5
1	37.17	100.67	86	133	66
2	38.33	97.33	82	134	66
⋮	⋮	⋮	⋮	⋮	⋮
39	51.25	77.67	55.5	153.5	71.5
40	52.42	74.33	51.5	154.5	71.5

Pearson

Product allocation Case 2 out 2



- A warehouse has 40 storage locations arranged in 4 conventional racks
- The optimization is Excel Solver gives the following solution

Storage location	Product 1	Product 2	Product 3	Product 4	Product 5
1	1	0	0	0	0
2	0	0	0	1	0
⋮	⋮	⋮	⋮	⋮	⋮
39	0	0	1	0	0
40	0	0	1	0	0

- The optimal cost is 2451.18.

Pearson
Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Warehousing Operations (1 of 9)

- Efficient and effective warehousing management can be an exacting task
- Workforce motivation can be difficult because the work can be:
  - Repetitive in nature
  - Strenuous and physically demanding
  - Occasionally dangerous

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Warehousing Operations (2 of 9)

- Some prominent operational issues include:
  - Warehousing productivity analysis
  - Safety considerations
  - Hazardous materials
  - Warehousing security
  - Cleanliness and sanitation

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Warehousing Operations (3 of 9)

- Warehousing productivity analysis
  - Productivity is a measure of output divided by input, e.g, cases picked and shipped per hour
  - Examples of representative measures include:
    - Cases shipped per person
    - Product lines shipped per person
    - Pallets shipped per person
    - Average warehouse capacity used
    - Forklift capacity used
  - Ways to improve productivity: 1) identify bottlenecks in the process, 2) develop methods to reduce and eliminate inefficiencies

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Warehousing Operations (4 of 9)

- Safety considerations
  - Warehousing facilities can be dangerous workplaces
    - Forklifts account for one out of every six workplace deaths[8]
  - Warehouse safety can by influenced by governmental regulations
    - In the U.S. safety standards have been set for warehousing equipment, and operations are enforced by Occupational Safety and Health Administration (OSHA)

[8]Eric Scharski, "Want a Safer Warehouse? Start with Forklift Users," *Food Logistics*, January/February 2015, 36–41.

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

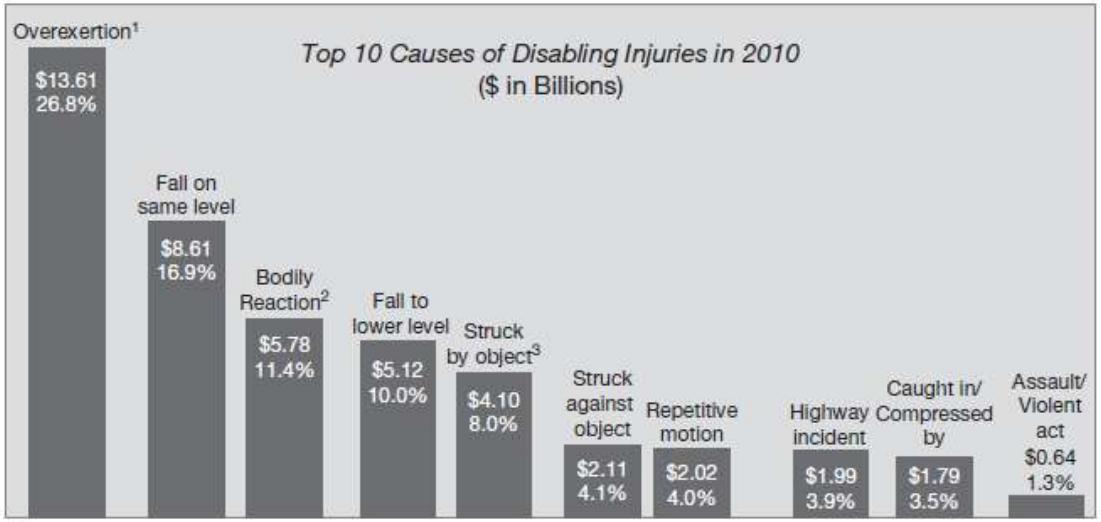
Warehousing Operations (5 of 9)

- Safety considerations
  - Warehouse safety categories include:
    - Employee
    - Property (e.g. from fire incidents)
    - Motor vehicles (e.g, forklift, tractors trailers)

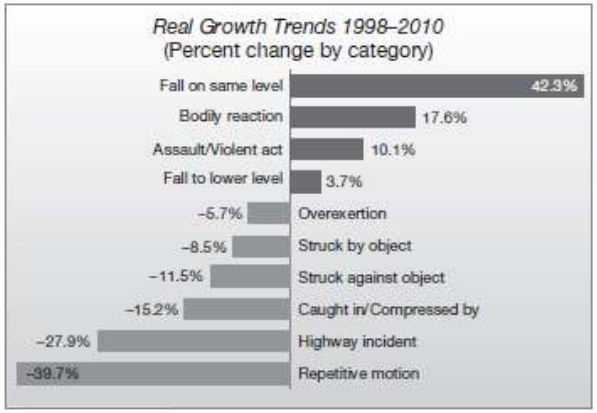
Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Figure 10.3: Workplace Safety Issues



[1]Overexertion – Injuries from excessive lifting, pushing, pulling, holding, carrying, throwing
[2]Bodily reaction – Injuries from bending, climbing, reaching, standing, sitting, slipping or tripping without falling
[3]Struck by object – Such as a tool falling on a worker from above
[4]Struck against object – Such as a worker walking into a door
[5]Repetitive motion – Injuries due to repeated stress or strain



Source: Reprinted with permission of Liberty Research Institute for Safety, August 30, 2012.

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Warehousing Operations (6 of 9)

- Hazardous materials (hazmat)
  - Examples include:
    - Explosives
    - Flammable liquids
    - Flammable solids
  - Government regulations often require shipping documents indicating the hazardous materials
  - Many countries also require the use of safety data sheets (SDS)

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Warehousing Operations (7 of 9)

- Warehousing security
  - Example potential threats include:
    - Theft
    - Pilferage
    - Heat and humidity
    - Vandalism
    - Fire
    - Loss of electricity

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Warehousing Operations (8 of 9)

- Warehousing security
  - Focuses on protecting products and preventing their theft
  - Can be enhanced by focusing on people, facilities, and processes

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Table 10.1: Possible Shortcomings in Warehousing Security

Shortcoming	Comment
Making it too easy for dock personnel to work in collusion with truck drivers	Fewer than 5% who commit crimes are prosecuted
Relying on safeguards that simply don't work	Security cameras aren't always turned on
Approach to theft is too reactive	Don't wait until theft reaches an "unacceptable" level
Not weeding out on-the-job substance abusers or dealers	Approximately 90% of drug users either steal or deal to support their habit
Not checking your checkers on a frequent basis	They may become complacent
Not making it easy to report theft and substance abuse	Outsourcing the reporting may be more effective
Hiring high-risk employees	An ounce of prevention is worth a pound of cure

Source: Julia Kuzeljevich, "The Seven Deadly Sins in Warehouse Security," *Canadian Transportation & Logistics*, April 2006, 44.

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Warehousing Operations (9 of 9)

- Cleanliness and sanitation issues
  - Particularly important in many industries, such as the foodservice industry
  - Can have a positive impact on employee safety, morale, and productivity while also reducing employee turnover[9]
  - Require common sense and due diligence

[9]Ned Bauhof, "Keeping It Clean," *Beverage World*, July 2007, 77.

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Key Terms (1 of 2)

- Accumulating (bulk-making)
- Allocating (bulk-breaking)
- Assorting
- Contract warehousing
- Cross-docking
- Distribution center
- Dunnage
- Fixed slot location
- Fulfillment centers
- Hazardous material(s)
- Multiclient warehousing
- Occupational Safety and Health Administration (OSHA)

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.

Key Terms (2 of 2)

- Private warehousing
- Public warehousing
- Regrouping function
- Sorting out
- Throughput
- Variable slot location
- Velocity slotting
- Warehouse automation
- Warehouses
- Warehousing

Pearson

Copyright © 2018 Pearson Education, Ltd. All Rights Reserved.